

# Eigenvector Comparison: MNIST '0' vs EMNIST 'O'

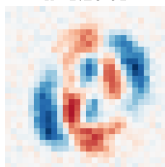
**Positive Eigenvalues**

**Negative Eigenvalues**

MNIST '0'

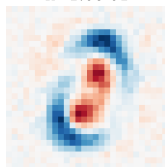
+1

$\lambda=1.2\text{e-}01$



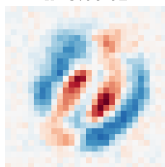
+2

$\lambda=1.0\text{e-}01$



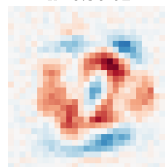
+3

$\lambda=9.0\text{e-}02$



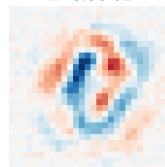
+4

$\lambda=6.9\text{e-}02$



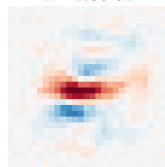
+5

$\lambda=5.9\text{e-}02$



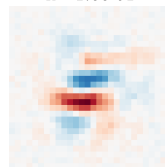
-1

$\lambda=-1.8\text{e-}01$



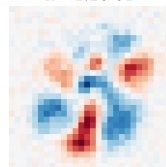
-2

$\lambda=-1.6\text{e-}01$



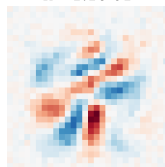
-3

$\lambda=-1.2\text{e-}01$



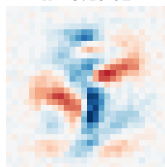
-4

$\lambda=-1.1\text{e-}01$



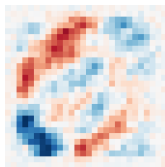
-5

$\lambda=-9.4\text{e-}02$

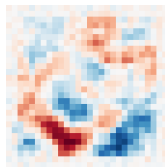


EMNIST 'O'

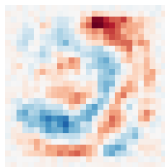
$\lambda=8.0\text{e-}02$



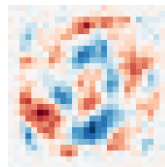
$\lambda=7.4\text{e-}02$



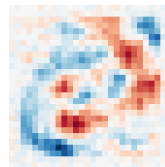
$\lambda=6.9\text{e-}02$



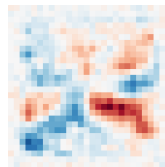
$\lambda=6.4\text{e-}02$



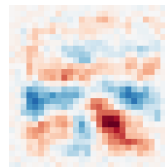
$\lambda=5.8\text{e-}02$



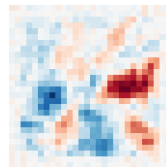
$\lambda=-1.5\text{e-}01$



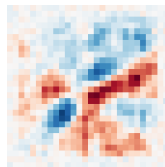
$\lambda=-1.3\text{e-}01$



$\lambda=-1.1\text{e-}01$



$\lambda=-9.8\text{e-}02$



$\lambda=-8.6\text{e-}02$

